Lab Sheet 2: DML Commands

INSERT, UPDATE and DELETE — SQL Answers with Expected Outputs

SQL dialect used: MySQL 8.x

The examples below use a consistent sample dataset. The outputs are representative results from MySQL after executing each operation. For clarity, each task is treated as a suitable standalone example rather than requiring the exact execution order of every previous task.

1. Insert one record into the Student table.

```
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (101, 'Aarav', 20, 'BCA', 'Meerut');
```

Expected output:

1 row inserted.

Student_ID	Name	Age	Course	City
101	Aarav	20	BCA	Meerut

2. Insert five student records using separate INSERT statements.

```
INSERT INTO Student VALUES (102, 'Riya', 19, 'BSc', 'Delhi');
INSERT INTO Student VALUES (103, 'Karan', 21, 'BCA', 'Meerut');
INSERT INTO Student VALUES (104, 'Neha', 20, 'BTech', 'Jaipur');
INSERT INTO Student VALUES (105, 'Vivek', 22, 'BCA', 'Delhi');
INSERT INTO Student VALUES (106, 'Priya', 19, 'BSc', 'Roorkee');
```

Expected output:

5 rows inserted. Each INSERT statement reports 1 row affected.

Student_ID	Name	Age	Course	City
102	Riya	19	BSc	Delhi
103	Karan	21	BCA	Meerut
104	Neha	20	BTech	Jaipur
105	Vivek	22	BCA	Delhi
106	Priya	19	BSc	Roorkee

3. Insert multiple records into the Student table using a single statement.

```
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES
(107, 'Ankit', 20, 'BCA', 'Delhi'),
(108, 'Simran', 21, 'BTech', 'Noida'),
(109, 'Rahul', 20, 'BSc', 'Meerut');
```

Expected output:

3 rows inserted in one statement.

Student_ID	Name	Age	Course	City
107	Ankit	20	BCA	Delhi
108	Simran	21	BTech	Noida
109	Rahul	20	BSc	Meerut

4. Insert records into the Employee table.

```
INSERT INTO Employee (Emp_ID, Emp_Name, Department, Salary)
VALUES
(201, 'Amit', 'IT', 45000),
(202, 'Sneha', 'HR', 40000),
(203, 'Rohan', 'Finance', 50000);
```

Expected output:

3 rows inserted.

Emp_ID	Emp_Name	Department	Salary
201	Amit	IT	45000.00
202	Sneha	HR	40000.00
203	Rohan	Finance	50000.00

5. Insert records into the Department table.

```
INSERT INTO Department (Dept_ID, Dept_Name, Location)
VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Jaipur'),
(3, 'Finance', 'Noida');
```

Expected output:

3 rows inserted.

Dept_ID	Dept_Name	Location
1	IT	Delhi
2	HR	Jaipur
3	Finance	Noida

6. Insert records into the Product table.

```
INSERT INTO Product (Product_ID, Product_Name, Price, Quantity)
VALUES
(301, 'Laptop', 55000, 10),
(302, 'Mouse', 800, 25),
(303, 'Keyboard', 1500, 15);
```

Expected output:

3 rows inserted.

Product_ID	Product_Name	Price	Quantity
301	Laptop	55000.00	10
302	Mouse	800.00	25
303	Keyboard	1500.00	15

7. Insert records into the Customer table.

```
INSERT INTO Customer (Customer_ID, Customer_Name, Email, Phone)
VALUES
(401, 'Ravi', 'ravi@example.com', '9876543210'),
(402, 'Pooja', 'pooja@example.com', '9876543211'),
(403, 'Mohit', 'mohit@example.com', '9876543212');
```

Expected output:

3 rows inserted.

Customer_ID	Customer_Name	Email	Phone
401	Ravi	ravi@example.com	9876543210
402	Pooja	pooja@example.com	9876543211
403	Mohit	mohit@example.com	9876543212

8. Insert records into the Course table.

```
INSERT INTO Course (Course_ID, Course_Name, Duration, Fees)
VALUES
(501, 'BCA', '3 Years', 90000),
(502, 'BSc', '3 Years', 75000),
(503, 'BTech', '4 Years', 160000);
```

Expected output:

3 rows inserted.

Course_ID	Course_Name	Duration	Fees
501	BCA	3 Years	90000.00

Course_ID	Course_Name	Duration	Fees
502	BSc	3 Years	75000.00
503	BTech	4 Years	160000.00

9. Insert a student record with values for all columns.

```
INSERT INTO Student
VALUES (110, 'Rakesh', 21, 'BCA', 'Haridwar');
```

Expected output:

1 row inserted.

Student_ID	Name	Age	Course	City
110	Rakesh	21	BCA	Haridwar

10. Insert a student record by specifying only selected columns.

```
INSERT INTO Student (Student_ID, Name, Course)
VALUES (111, 'Megha', 'BSc');
```

Expected output:

1 row inserted. Age and City receive NULL because they were not specified.

Student_ID	Name	Age	Course	City
111	Megha	NULL	BSc	NULL

11. Update the city of a particular student.

```
UPDATE Student
SET City = 'Delhi'
WHERE Student_ID = 101;
```

Expected output:

1 row affected.

Student_ID	Name	City
101	Aarav	Delhi

12. Update the salary of a particular employee.

```
UPDATE Employee
SET Salary = 48000
WHERE Emp_ID = 201;
```

Expected output:

1 row affected.

Emp_ID	Emp_Name	Salary
201	Amit	48000.00

13. Increase the salary of all employees by 10%.

```
UPDATE Employee
SET Salary = Salary * 1.10;
```

Expected output:

3 rows affected.

Emp_ID	Emp_Name	Salary
201	Amit	52800.00
202	Sneha	44000.00
203	Rohan	55000.00

14. Increase the price of all products by 5%.

```
UPDATE Product
SET Price = Price * 1.05;
```

Expected output:

3 rows affected.

Product_ID	Product_Name	Price
301	Laptop	57750.00
302	Mouse	840.00
303	Keyboard	1575.00

15. Change the course of a particular student.

```
UPDATE Student
SET Course = 'BTech'
WHERE Student_ID = 103;
```

Expected output:

1 row affected.

Student_ID	Name	Course
103	Karan	BTech

16. Update the department of an employee.

```
UPDATE Employee
SET Department = 'IT'
WHERE Emp_ID = 203;
```

Expected output:

1 row affected.

Emp_ID	Emp_Name	Department
203	Rohan	IT

17. Update the city of all students from Meerut to Delhi.

```
UPDATE Student
SET City = 'Delhi'
WHERE City = 'Meerut';
```

Expected output:

3 rows affected in the sample data.

Student_ID	Name	City
101	Aarav	Delhi
103	Karan	Delhi
109	Rahul	Delhi

18. Update the fees of a particular course.

```
UPDATE Course
SET Fees = 95000
WHERE Course_ID = 501;
```

Expected output:

1 row affected.

Course_ID	Course_Name	Fees
501	BCA	95000.00

19. Update multiple columns of a student simultaneously.

```
UPDATE Student
SET Age = 21,
Course = 'BTech',
City = 'Noida'
WHERE Student_ID = 104;
```

Expected output:

1 row affected.

Student_ID	Name	Age	Course	City
104	Neha	21	BTech	Noida

20. Delete a particular student record using Student_ID.

```
DELETE FROM Student
WHERE Student_ID = 111;
```

Expected output:

1 row deleted.

Deleted Student_ID
111

21. Delete all students belonging to a particular city.

```
DELETE FROM Student
WHERE City = 'Noida';
```

Expected output:

2 rows deleted in the sample data.

Deleted Student_ID	City
104	Noida
108	Noida

22. Delete employees whose salary is below a specified amount.

```
DELETE FROM Employee
WHERE Salary < 45000;
```

Expected output:

1 row deleted in the sample data (after Task 13).

Deleted Emp_ID	Emp_Name	Salary
202	Sneha	44000.00

23. Delete products whose quantity is zero.

```
DELETE FROM Product
WHERE Quantity = 0;
```

Expected output:

0 rows deleted because none of the sample products has Quantity = 0.

Result
No matching records

24. Delete records satisfying multiple conditions using AND/OR.

```
DELETE FROM Student
WHERE Age > 21
AND (Course = 'BCA' OR Course = 'BSc');
```

Expected output:

2 rows deleted in the sample data.

Deleted Student_ID	Age	Course
105	22	BCA
106	19	BSc

25. Complete Sequence of INSERT, UPDATE and DELETE

The following example demonstrates the complete DML sequence on a small Student table. The table is displayed after each operation so the effect of INSERT, UPDATE and DELETE is clear.

Initial table

```
CREATE TABLE Student_DML (
Student_ID INT PRIMARY KEY,
Name VARCHAR(50),
Age INT,
Course VARCHAR(50),
City VARCHAR(50)
);
```

Empty table created.

Student_ID	Name	Age	Course	City

After INSERT

```
INSERT INTO Student_DML
VALUES
(1, 'Aarav', 20, 'BCA', 'Meerut'),
(2, 'Riya', 19, 'BSc', 'Delhi'),
(3, 'Karan', 21, 'BTech', 'Jaipur');
```

3 rows inserted.

Student_ID	Name	Age	Course	City
1	Aarav	20	BCA	Meerut
2	Riya	19	BSc	Delhi
3	Karan	21	BTech	Jaipur

After UPDATE

```
UPDATE Student_DML
SET City = 'Delhi', Age = 21
WHERE Student_ID = 1;
```

1 row affected.

Student_ID	Name	Age	Course	City
1	Aarav	21	BCA	Delhi
2	Riya	19	BSc	Delhi
3	Karan	21	BTech	Jaipur

After DELETE

```
DELETE FROM Student_DML
WHERE Student_ID = 2;
```

1 row deleted.

Student_ID	Name	Age	Course	City
1	Aarav	21	BCA	Delhi
3	Karan	21	BTech	Jaipur

Quick DML reference

Command	Purpose	Example
INSERT	Add new records	INSERT INTO Student VALUES (...);
UPDATE	Modify existing records	UPDATE Student SET City='Delhi' WHERE Student_ID=1;
DELETE	Remove selected records	DELETE FROM Student WHERE Student_ID=1;

Note: Exact MySQL client messages can differ slightly between MySQL Workbench and the command-line client. The tables show the expected data/result after the stated operation.